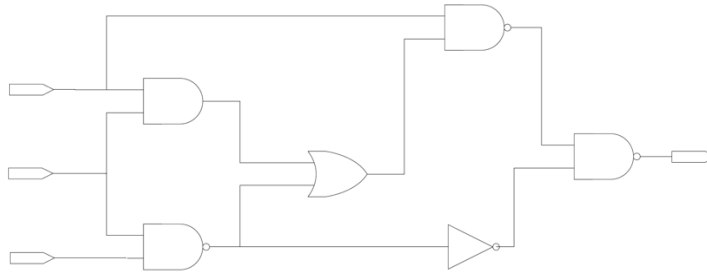


## Homework # 2

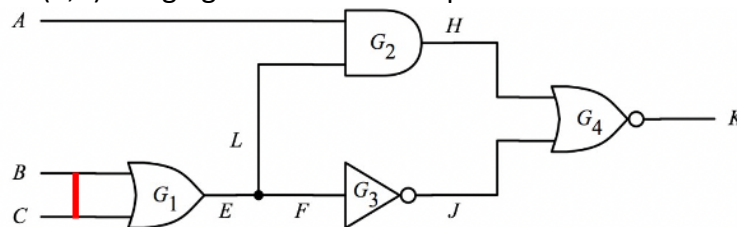
## Solution

1. For the logic circuit in figure 1, Compute a) the total number of single stuck-at faults b) the total number of stuck-open faults c) the total number of all possible multiple stuck-at fault combinations (assume 2-inputs NANDs and NORs use 4 Transistors, 2-input ANDs and ORs use 6 transistors, and NOTs use 2 transistors).



- a)  $N = \text{Number of fault sites} = \text{PIs} + \text{gates} + \text{fanout branches} = 15$   
 Therefore, Number of single faults =  $15 \times 2 = 30$
- b) Total number of transistors =  $6 \times 2 + 4 \times 3 + 2 = 26$
- c) Number of single and multiple faults =  $3^N - 1 = 3^{15} - 1 = 14348906$   
 Number of multiple stuck-at-faults (not including single stuck-at-faults) =  $3^N - 2N - 1 = 14348876$

2. Fill in the table with (B,C) bridging fault. Which test pattern detects a wired-OR fault?



No pattern can detect a wired-OR gate.

| Inputs<br>A B C | Fault-free<br>Output | Wired-<br>OR | Wired-<br>AND | B-<br>Dominant |
|-----------------|----------------------|--------------|---------------|----------------|
| 0 0 0           | 0                    | 0            | 0             | 0              |
| 0 0 1           | 1                    | 1            | <u>0</u>      | <u>0</u>       |
| 0 1 0           | 1                    | 1            | <u>0</u>      | 1              |
| 0 1 1           | 1                    | 1            | 1             | 1              |
| 1 0 0           | 0                    | 0            | 0             | 0              |
| 1 0 1           | 0                    | 0            | 0             | 0              |
| 1 1 0           | 0                    | 0            | 0             | 0              |
| 1 1 1           | 0                    | 0            | 0             | 0              |

3. Fill in the table with a Wired-AND feedback bridging fault (x, z).

| Inputs<br>X Y | Fault-free<br>Output | Faulty<br>Output |
|---------------|----------------------|------------------|
| 0 0           | 0                    | 0                |
| 0 1           | 1                    | 0                |
| 1 0           | 1                    | <u>Unchanged</u> |
| 1 1           | 1                    | 1                |

